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Instruction

What the question is

From the Previous HW-Since I did something similar

PDA'S & Pumping Lemma from PDAs and Turing Machines

Pumping Lemma Practice: 2.31 (hint: use $s=0p12p0p$), 2.32 (NEW)

2.31 Let B be the language of all palindromes over $\{0,1\}$ containing equal numbers of 0s and 1s. Show that B is not context free.

This is from the previous assignment that was omitted:

2.31 Let B be lang for all palindrom over $\{0,1\}$ containing the
 $\#$'s of 0's & 1's. To prove B is not CFL by using contradiction.
 Assume that B is a CFL. Pumping Lemma = P.L

Since B is a CFL, then by P.L, there's a $\#$ p (the P. length
 where, if z is any str in B of length at least p , then z may
 be / in 5 pieces (or 3) $z = uvxyz$ w/ these condit:

$$1. |v| \geq 1 \text{ or } |x| \geq 1 \quad 3. i \geq 0, uv^i xy^i z \in A$$

(Not a part of z)

$$2. |vxy| \leq p$$

z is a mem of B of length at least p . $n=2$ str z Assume
 $z = 0^2 1^4 0^2$ $z = 00111100$ can be / into 5 by: val

$$\frac{00}{u} \frac{11}{v} \frac{1}{x} \frac{10}{y} \frac{0}{z}$$

Assume the str

$$001111110100 = z'$$

obtained str z' is not a palindrome
 after applying 3rd cond of
 P.L $z' \notin B$.

Lang B not following cond 3
 of P.L

Apply 3rd cond. for $i=2$

$$\frac{00}{u} \left(\frac{11}{v}\right)^2 \frac{1}{x} \left(\frac{10}{y}\right)^2 \frac{0}{z}$$

$$\frac{00}{u} \left(\frac{1111}{v}\right) \frac{1}{x} \left(\frac{1010}{y}\right) \frac{0}{z}$$

$z = 0^n 1^{2n} 0^n$ z is a mem of B of length at least p

Assume val of $n=2$ for str z . $z = 0^2 1^4 0^2$ $z = 00111100$
 / into $uvxyz$

$$\frac{0}{u} \frac{01}{v} \frac{11}{x} \frac{1}{y} \frac{00}{z}$$

$$\frac{0}{u} \left(\frac{01}{v}\right)^2 \frac{11}{x} \left(\frac{1}{y}\right)^2 \frac{00}{z}$$

$$\frac{0}{u} \left(\frac{0101}{v}\right) \frac{11}{x} \left(\frac{11}{y}\right) \frac{00}{z}$$

3rd cond of P.L $i=2$

Assume str $0010111100 = z'$

z' Not Palindrome after apply 3rd cond of P.L &
 $z' \notin B$. Pumped str. the # of 0's & 1's $\neq$. Thus, $B \notin$ CFL
 Thus, two tries clearly prove that B is not CFL.

- 2.32 Let $\Sigma = \{1, 2, 3, 4\}$ and $C = \{w \in \Sigma^* \mid \text{in } w, \text{ the number of 1s equals the number of 2s, and the number of 3s equals the number of 4s}\}$. Show that C is not context free.

2.32 | let's say the oppo. C is context free. so, C has pumping length p .

$$s = 1^p 3^p 2^p 4^p \in C \text{ w/ } |s| \geq p$$

$uVXYZ$ exists such that

a) $uv^i xy^i z \in C$ for all $i \geq 0$

b) $|VY| > 0$

c) $|VXY| \leq p$

Prove by contradiction, no matter what val $uVXYZ$

C1: if VXY has a 1 = $uv^2 xy^2 z \notin C$. b/c it can't be the same # of 1's & 2's. hence c). VXY can't have any 2's.

C2: If VXY has a 2

VXY can't have any 1's

C3: If VXY has a 3

VXY can't have any 4's

C4: If VXY has a 4

VXY can't have any 3's

Thus, A, B equation are contradicting. you see C is not CFL.

This is from the previous assignment that was omitted:

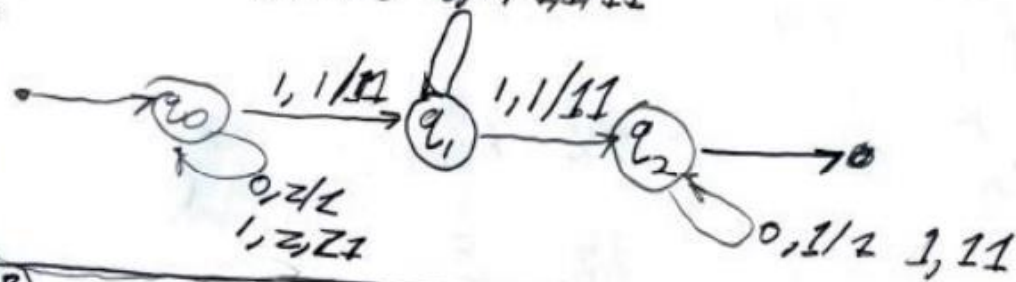
2.4 Give context-free grammars that generate the following languages. In all parts, the alphabet Σ is $\{0,1\}$.

- ^Aa. $\{w \mid w \text{ contains at least three 1s}\}$
- b. $\{w \mid w \text{ starts and ends with the same symbol}\}$
- c. $\{w \mid \text{the length of } w \text{ is odd}\}$
- ^Ad. $\{w \mid \text{the length of } w \text{ is odd and its middle symbol is a 0}\}$
- e. $\{w \mid w = w^{\mathcal{R}}, \text{ that is, } w \text{ is a palindrome}\}$
- f. The empty set

2.4) A)

$S \rightarrow X1X1X$

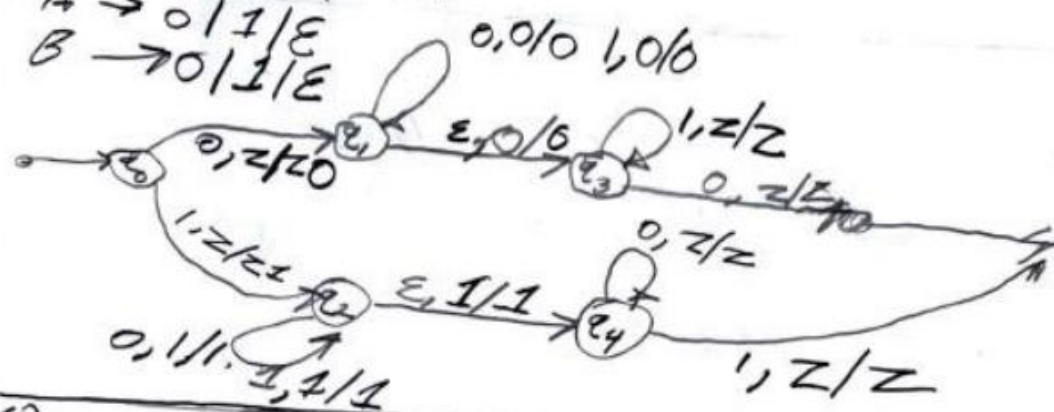
$X \rightarrow 0111 \epsilon$



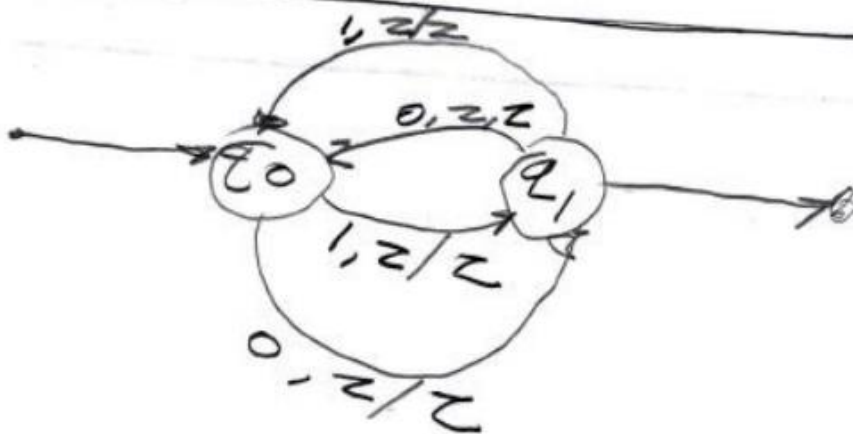
B) $S \rightarrow 0A011B1$

$A \rightarrow 011 \epsilon$

$B \rightarrow 011 \epsilon$



C)

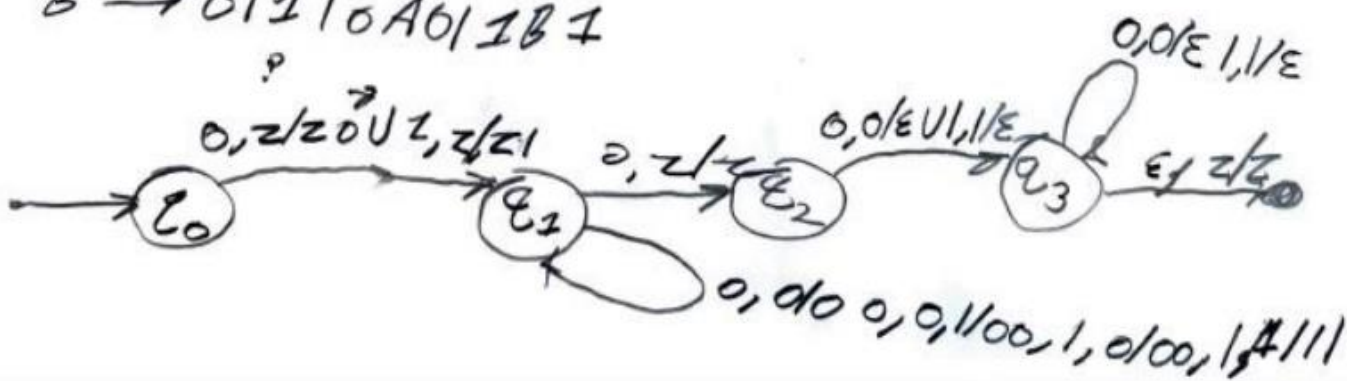


$S \rightarrow 0110A011B1$
 $A \rightarrow 0110A011B1$
 $B \rightarrow 0110A011B1$

D) $S \rightarrow 01110A011B1$

$A \rightarrow 01110A011B1$

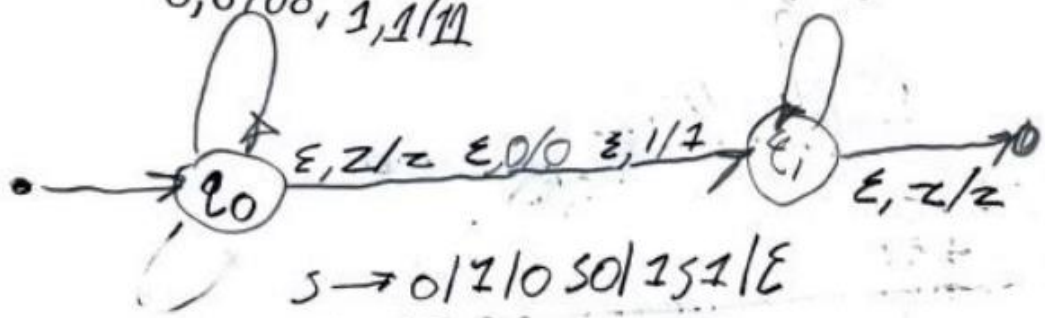
$B \rightarrow 01110A011B1$



E)

$0, z/0$
 $1, z/1$
 $0, 0/0$
 $1, 1/1$

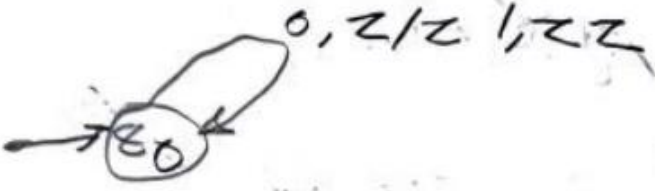
$0, 0/\epsilon$
 $1, 1/\epsilon$

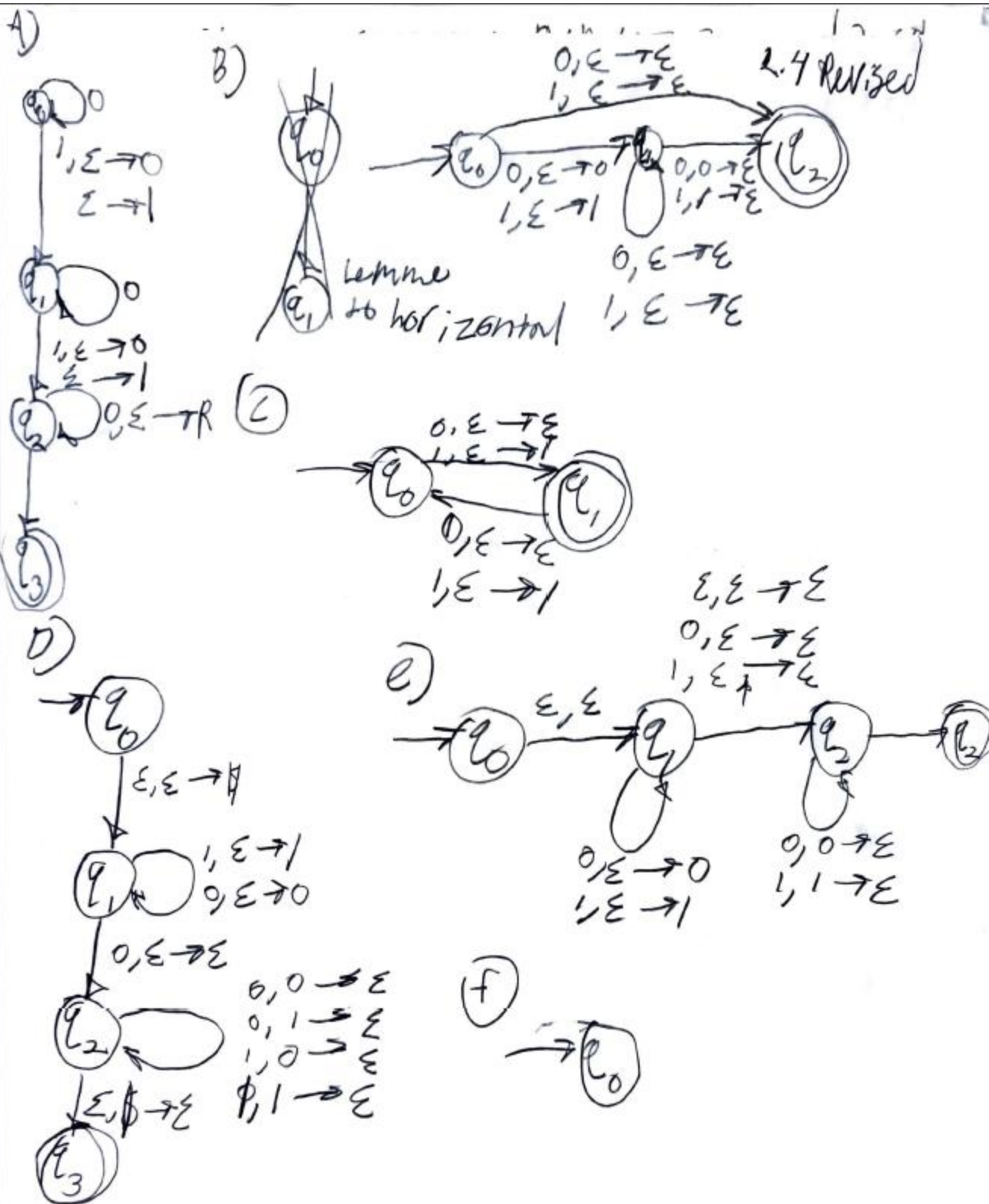


$S \rightarrow 0/1/0/1/1/\epsilon$

F)

$S \rightarrow X$
 $X \rightarrow YZ$
 $Y \rightarrow \epsilon$
 $Z \rightarrow 0/1$





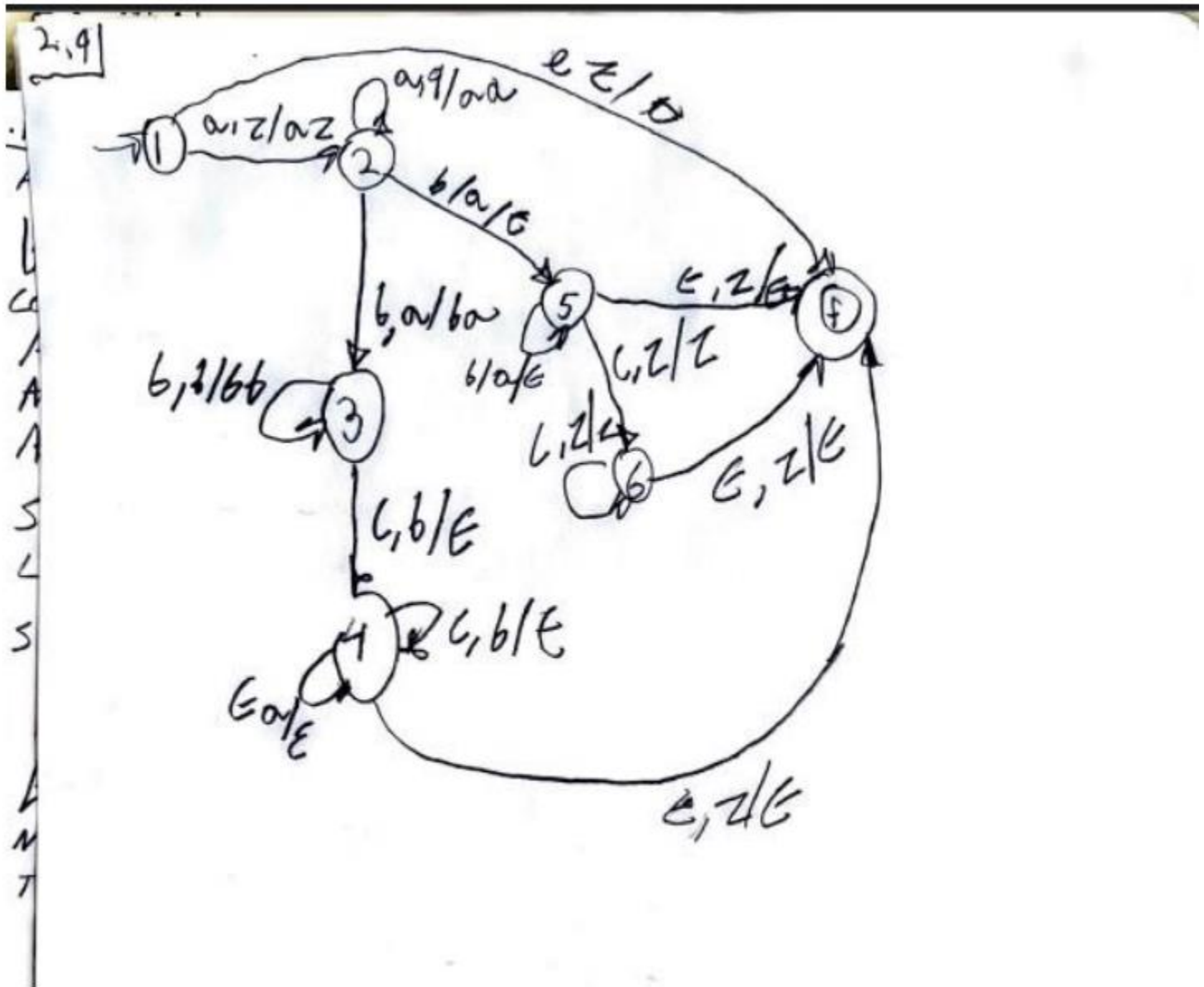
Build state diagram of PDA to recognize language of 2.9 (no big loop)

This is from the previous assignment that was omitted:

2.9 Give a context-free grammar that generates the language

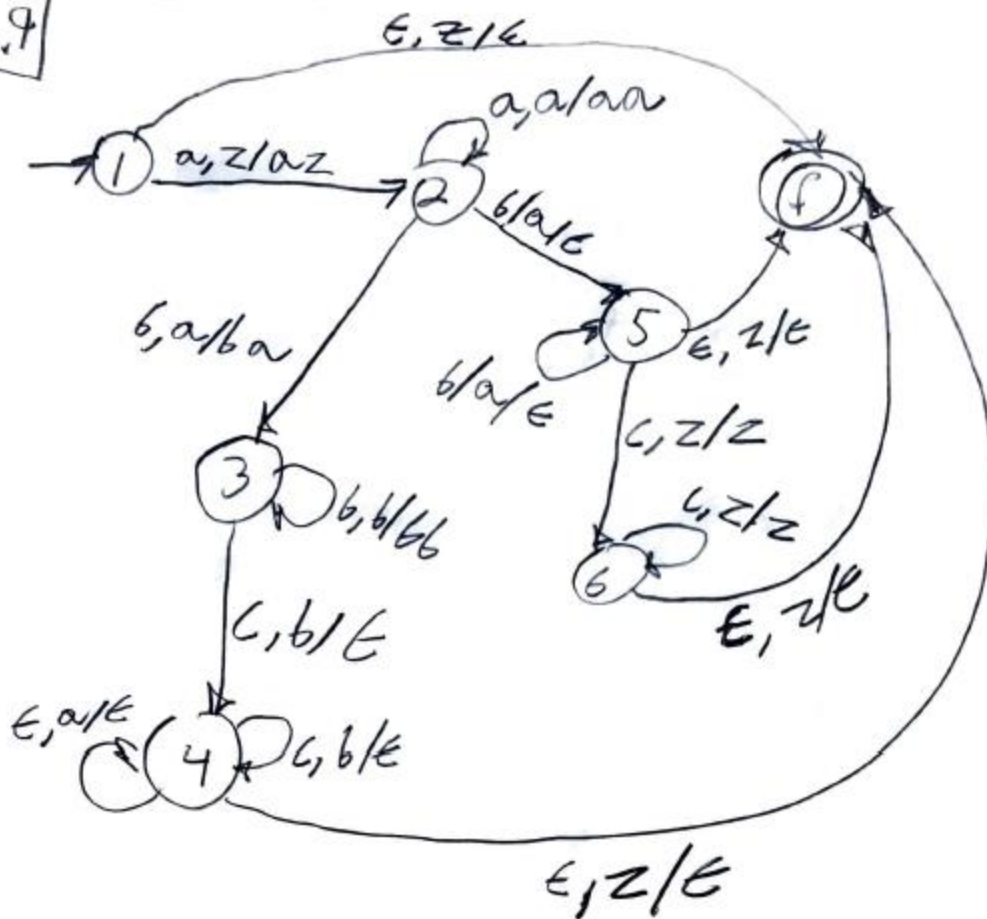
$$A = \{a^i b^j c^k \mid i = j \text{ or } j = k \text{ where } i, j, k \geq 0\}.$$

Is your grammar ambiguous? Why or why not?



Redid it since I always forget some tiny detail because I think it's there.

2.9



Build state diagram of PDA to recognize language of 2.6b (no big loop)

2.6 Give context-free grammars generating the following languages.

a. The set of strings $\{a^n b^n \mid n \geq 0\}$ with one a^n before the b^n

b. The complement of the language $\{a^n b^n \mid n \geq 0\}$

a. $\{a^n b^m \mid n \neq m\}$

a. $\{a^n b^m \mid n \neq m\} = \{a^n b^m \mid n < m\} \cup \{a^n b^m \mid n > m\}$

2.68) comp of lang $\{a^n b^n \mid n \geq 0\}$

Let L be the lang that is a complement of given Lang. L can be obtained as $L = \{a^n b^m \mid n \neq m\} \cup \{(a \cup b)^* b a (a \cup b)^*\}$

Let's do this

$$L_1 = \{a^n b^m \mid n \neq m\}$$

$$L_2 = \{(a \cup b)^* b a (a \cup b)^*\}$$

The context-free grammar generating the lang L_1 is

$$S_1 \rightarrow a S_1 b \mid T \mid U$$

$$T \rightarrow a T \mid a$$

$$U \rightarrow U b \mid a$$

The context-free grammar generating the lang $L_2 =$

$$S_2 \rightarrow R b a R$$

$$R \rightarrow R R \mid a \mid b \mid \epsilon$$

Therefore, the required context-free grammar generating the lang L is given by

$$L = L_1 \cup L_2$$

$$S \rightarrow S_1 \mid S_2$$

$$S_1 \rightarrow a S_1 b \mid T \mid U$$

$$S_2 \rightarrow R b a R$$

$$T \rightarrow a T \mid a$$

$$U \rightarrow U b \mid a$$

$$R \rightarrow R R \mid a \mid b \mid \epsilon$$

The complement of Lang $\{a^n b^n \mid n \geq 0\}$

2.68
Revised

CFG:

$$L' = \{w \mid w = a^n b^n, n \geq 0\}$$

$$\text{Let } G = (V, \Sigma, S, R)$$

$$V = \{S, A, B, C, D\}$$

$$\Sigma = \{a, b\}$$

$$CFG = S \rightarrow A | aB | cb$$

$$A \rightarrow aA | Aa | Ab | bA | ba$$

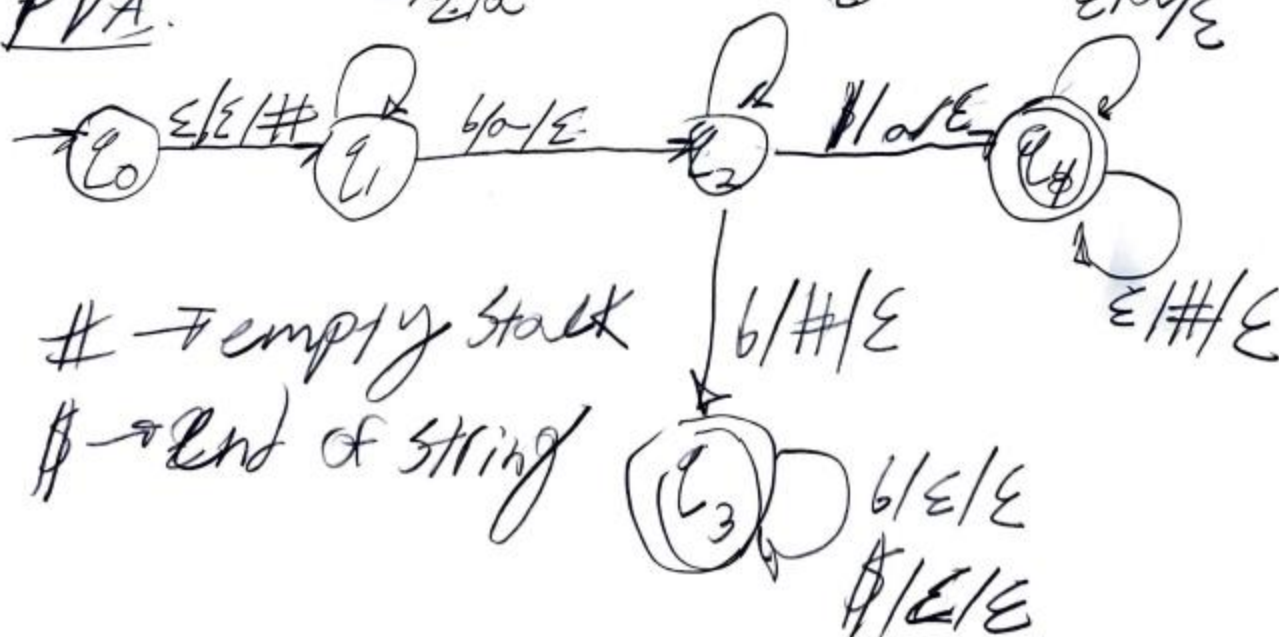
$$B \rightarrow D | aB$$

$$C \rightarrow D | cb$$

$$D \rightarrow aDb | \epsilon$$

$$L = \{aab, abb, \dots\}$$

PDA:



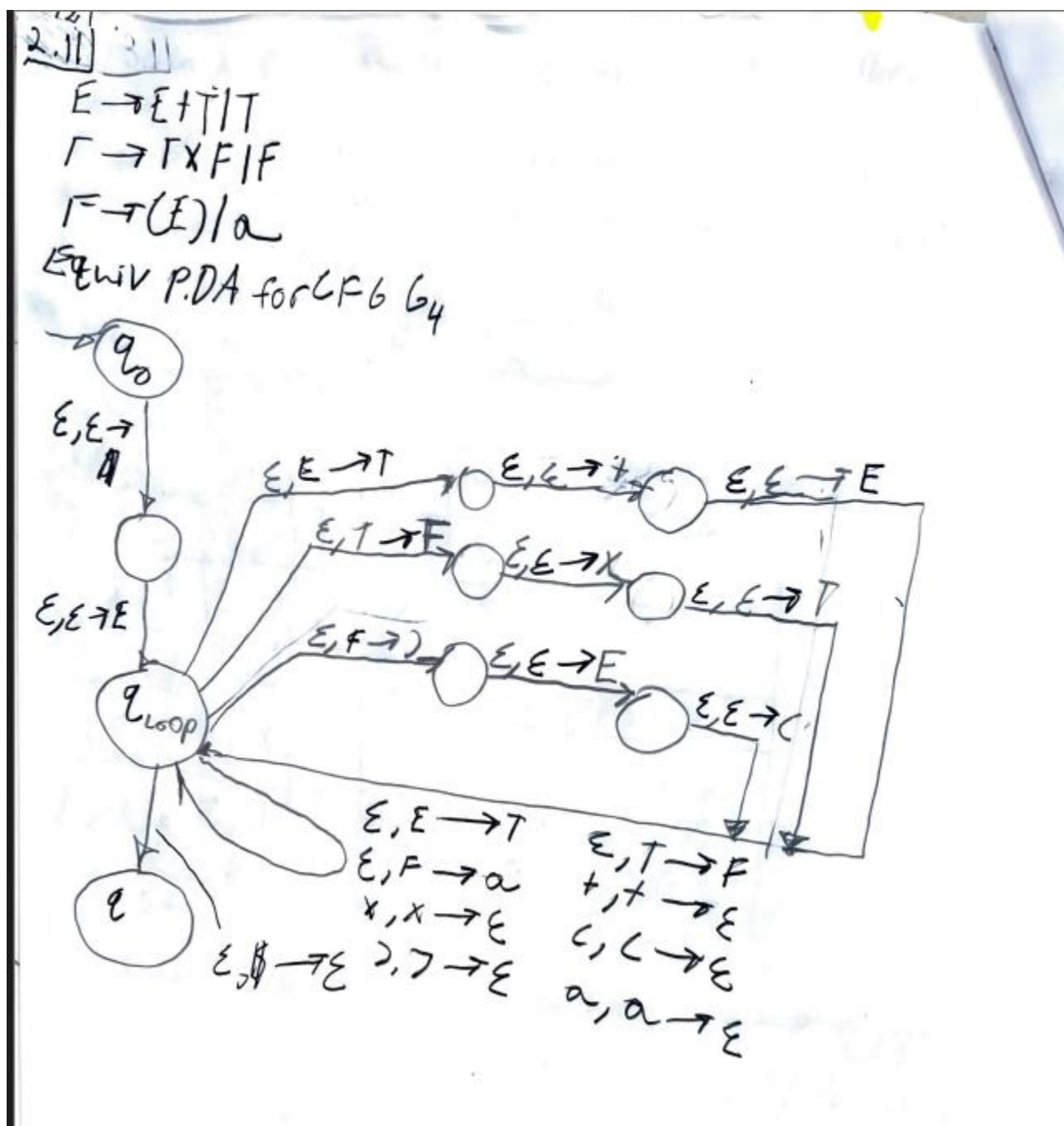
2.11 Convert the CFG G_4 given in Exercise 2.1 to an equivalent PDA, using the procedure given in Theorem 2.20.

2.1 Recall the CFG G_4 that we gave in Example 2.4. For convenience, let's rename its variables with single letters as follows.

$$\begin{aligned} E &\rightarrow E + T \mid T \\ T &\rightarrow T \times F \mid F \\ F &\rightarrow (E) \mid a \end{aligned}$$

Give parse trees and derivations for each string.

- | | |
|--------|----------|
| a. a | c. a+a+a |
| b. a+a | d. ((a)) |



- ^A2.18
- a. Let C be a context-free language and R be a regular language. Prove that the language $C \cap R$ is context free.
 - b. Let $A = \{w \mid w \in \{a, b, c\}^* \text{ and } w \text{ contains equal numbers of } a\text{'s, } b\text{'s, and } c\text{'s}\}$. Use part (a) to show that A is not a CFL.

10 A) We have a CFL L & a Reg Lang R , & we want to show that their intersection, $L \cap R$, is also context-free. Since L is a CFL, we know that there exists a pushdown automaton (PDA) M_1 that recognizes L . Similarly, since R is a reg lang, there exists a DFA, M_2 that recognizes R .

2. To prove that $L \cap R$ is CFL, we need to construct PDA, say M , that recognizes it. The idea is to use the PDA M_1 to keep track of the context-free part of the reg lang R .

Let M_1 recognize L , where $M_1 = (Q_1, \Sigma, \Gamma, \delta_1, q_1, F_1)$

Let M_2 recognize R , where $M_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$

We construct M to recognize $L \cap R$, where $M = (Q, \Sigma, \Gamma, \delta, q, F)$ like:

1. $Q = \{ (q_1, q_2) \mid q_1 \in Q_1 \text{ & } q_2 \in Q_2 \}$ represents the states of the combined PDA.
2. Σ is assumed to be the same in M_1 & M_2 .
3. $\Gamma = \Gamma$ represents the stack symbols of the PDA.
4. δ is defined as follows: each $(q_1, q_2) \in Q$, each $a \in \Sigma$, & each $b \in \Gamma$, let $\delta((q_1, q_2), a, b) = \delta_1(q_1, a, b) \text{ & } \delta_2(q_2, a)$ means that for a given input sym a , M_1 will move q_1 a new state using its transit. funct δ_1 & push or pop sym from stack. Then M_2 will move to a new state using its transit. funct δ_2 , & determine whether remain input belongs to R .
5. $q = (q_1, q_2)$ is the start state of M .
6. $F = \{ (q_1, q_2) \mid q_1 \in F_1 \text{ & } q_2 \in F_2 \}$ is the start states of M .

by construction, M recog. Lang $L \cap R$. Therefore, we have shown that the intersection of a context-free L & reg lang is context-free.

2.18B

Assume for the contra. that A is a CFL. then from Part A) we know that the lang $A \cap R$ is also CF for any reg lang R .

Let $B = \{a^n b^n c^{2/n} \mid n \geq 0\}$, which we know is not C.F. ~~we can see that~~ we can see that B & A have similar structure, as they both require $\approx$ #'s of a's, b's & c's. Let $R = a^* b^* c^*$, which is a reg lang & has the property of ordering the chars a, b, & c.

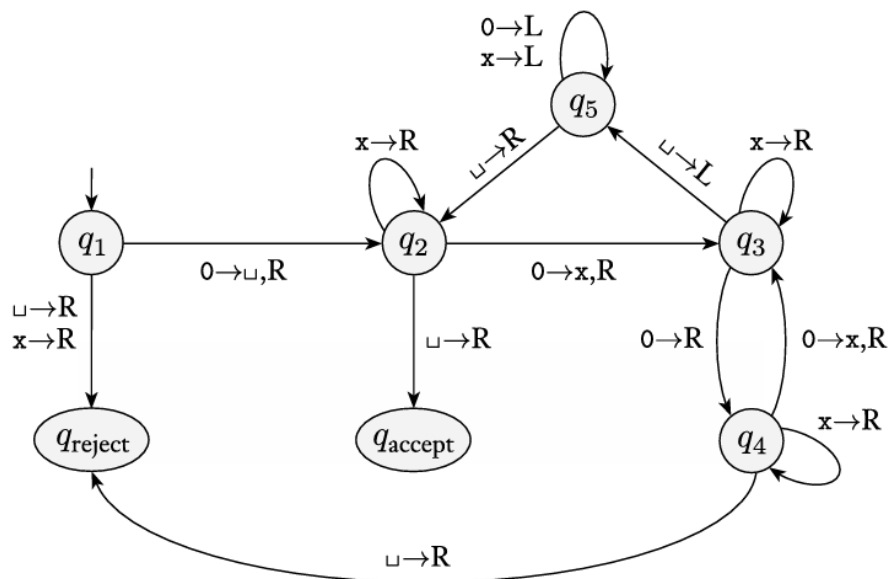
then we have $A \cap R = \{a^n b^n c^{2/n} \mid n \geq 0\} = B$. If A were C.F, then $A \cap R$ would also be C.F. But since B is not C.F, we have contra. Thus, our assumption that A is a CFL = fals. A is not a CFL.

///OMITTED QUESTIONS THAT I ANSWERED/ISH:

TM Configurations: 3.1b, 3.2b, 3.2c

3.1 This exercise concerns TM M_2 , whose description and state diagram appear in Example 3.7. In each of the parts, give the sequence of configurations that M_2 enters when started on the indicated input string.

Ab. 00.



Have 2 run machine M_2 on input 00. The starting Config = $q_1, 00$. The seq of Config that the machine enters when started on the input 00 is as follows.

	At:	→:	Go to the
$q_1, 00$	$q_1, 0$	R	q_2
$\sqcup q_2, 0$	$q_2, 0$	R	q_3
$\sqcup x q_3, \sqcup$	$q_3, \sqcup$	L	q_5
$\sqcup \sqcup q_5, x \sqcup$	q_5, x	L	q_5
$q_5 \sqcup x \sqcup$	q_5, x	R	q_2
$\sqcup q_2, x \sqcup$	$q_5, \sqcup$	R	accept state
$\sqcup x q_2, \sqcup q_2, x$		R	accept state
$\sqcup x \sqcup q_2, \sqcup$			
accept state			

~~The state q_5 on the machine goes to q_2 and~~
 I think I don't need to over explain it.
 Finally M_2 enters $q_{\text{accept state}}$ & is accepted.

3.2 This exercise concerns TM M_1 , whose description and state diagram appear in Example 3.9. In each of the parts, give the sequence of configurations that M_1 enters when started on the indicated input string.

a. ~~1#1~~

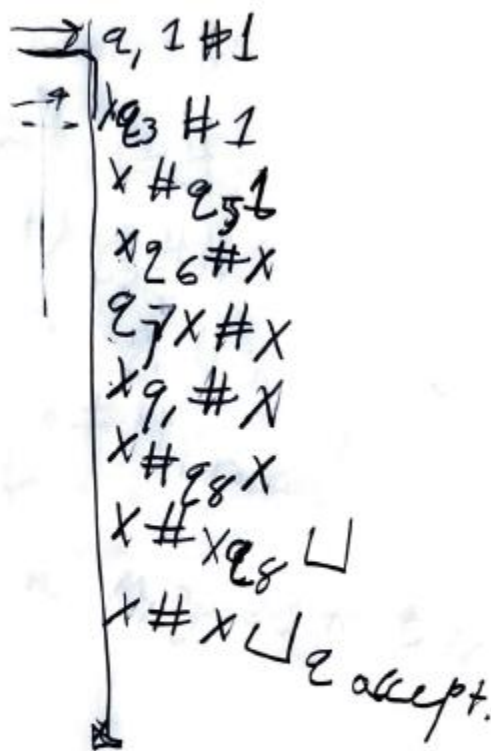
b. 1#1.

c. 1##1.

~~1#1#1~~

~~1#1#1#1~~

3.26



M_1 enters q_{accept} state. Thus, input 1#1 accepts.

3.26 Let's do $1\#\#1$
 Sequence of config that M enters

$q_1 1\#\#1$

$\times q_3\#\#1$

$\times \#q_5\#1$

$\times \#\#q_{reject}1$

[q_5 is not reading $\#$, so it enters to reject state.]

Fin M goes to q_{reject} & so $1\#\#1 =$
 rejected.

TM trivia: 3.5

A3.5 Examine the formal definition of a Turing machine to answer the following questions, and explain your reasoning.

- Can a Turing machine ever write the blank symbol $\sqcup$ on its tape?
- Can the tape alphabet Γ be the same as the input alphabet Σ ?
- Can a Turing machine's head *ever* be in the same location in two successive steps?
- Can a Turing machine contain just a single state?

3.5 A) yes - T.M. can write blank sym. on its tape according to the def. A T.M. is a 7 tuple $\langle Q, \Sigma, \Gamma, \delta, q_0, q_{accept}, q_{reject} \rangle$ where $Q, \Sigma, \Gamma = \text{finite}$

Σ is the input alphabet not containing the blank sym.
 Γ is the tape alphabet. A T.M. can write any char. in Γ on its tape. T.M. can write blank sym. on its tape.

B) No, The Tape alphabet Γ never same as input alphabet doesn't contain blank sym. Always Σ is subset of Γ , but never same as Σ .

C) yes. T.M. head be in the same location in 2 successive steps. When the T.M. head is left of the end of tape. If T.M. again try to move left side then T.M. head in same loc as the prev moves.

D) No,

They are distinct & independent

A T.M. is a 7 tuple $\langle Q, \Sigma, \Gamma, \delta, q_0, q_{accept}, q_{reject} \rangle$
 $Q = \text{set of states}$

q_{accept}, q_{rej} belongs Q but they DON'T = each other
 T.M. contains at least 2 states.

3.8 Give implementation-level descriptions of Turing machines that decide the following languages over the alphabet $\{0,1\}$.

~~a. $\{w \mid w \text{ contains an equal number of 0s and 1s}\}$~~

b. $\{w \mid w \text{ contains twice as many 0s as 1s}\}$

c. $\{w \mid w \text{ does not contain twice as many 0s as 1s}\}$